📚 SIT REPOSITORY - Comprehensive Documentation

Project Overview

The **SIT Repository** is a secure, full-stack academic platform developed for **Sethu Institute of Technology**. It serves as a central hub for faculty to archive research papers while providing administrators with high-level analytics on institutional research productivity.

1. The Home Page

The landing page serves as the public view (before login) of SIT REPOSITORY .It pulls from 7 data points to provide a total view:

- **Total Faculty Count:** Real-time count of active institutional staff.
- **Principal/Dean/HOD Distribution:** 3 separate counters showing leadership distribution.
- **Staff Tier Breakdown:** Count of Professors, Associate, and Assistant Professors.
- **Publication Totals:** Big counters for total Journals vs. Conferences.
- **Top 5 Publishers:** A leaderboard showing the most active researchers with their photos.
- **Recent Publishes:** The newest top 5 additions to the repository.
- **Departmental Analytics:** Organized bar charts (Department vs. Count) and growth lines (Year vs. Count) and department wise faculty count.

2. Authentication & The "Admin Gatekeeper" Logic

Your system is designed for high security through an approval-based workflow:

- **Create New Account:** When a user registers, they aren't "in" yet.
    - **The Notification:** A signal is sent to the Admin’s "Action Requests" panel.

    - **The Lock:** The user’s status is set to Pending in MongoDB. Any attempt to login before Admin approval results in an "Access Denied" or "Pending Approval" message.
- **Forgot Password (The 6-Digit Secret):**
    - **Verification:** The system sends an automated email to the user with a **6-figure secret number**.
    - **Recovery:** The user enters this code. Once verified, the code acts as a temporary key or allows a password reset, which is immediately updated in the user's metadata in the database.

3. The Dashboard & "Smart Redirect" Gateway

The Dashboard is the central hub, but the **Profile Picture** acts as a dynamic gateway:

- **Role-Based Redirection:**
    - **Normal User:** Clicking the profile pic opens their **Personal Profile Page**.
    - **Admin:** Clicking the profile pic is the only way to enter the **Admin Management Page**.
- **The Repository Feed:** Publications are sorted using **"Newest First" logic**. The system queries the database by timestamp, ensuring the most recent research always sits at the very top of the feed.
- **Advanced Filters:** Users can narrow down the entire repository by Author, Title, Domain, or Type (Journal/Conference).

4. The Faculty Profile Workspace

This is where the user manages their "Digital Academic Identity":

- **Publication Management:** All papers uploaded by the user are listed.
- **The profile details :** Here the detail about the user will be displayed.

- **The Deletion Request:** A user cannot just delete a paper. Clicking "Delete" sends a **Paper Deletion Request** to the Admin to prevent accidental or unauthorized data loss.
- **Credential Updates:**
    - **Position/Dept Change:** Requesting a change in rank (e.g., Asst. Prof to Assoc. Prof) or Department sends a specific request to the Admin.
    - **Self-Service:** Users can change their **Profile Picture** and **Password** directly without admin intervention.

5. The Upload Wizard & PDF Compression

This is the most technical part of your project:

- **Storage Logic:** Publications are stored with a link between the physical file on the server and the metadata in MongoDB.
- **The PDF Compression Engine:** To save server space and ensure fast loading for viewers, the system passes the uploaded PDF through a compression script **(Ghostscript)**. It reduces the file size while maintaining text readability before saving it to the permanent directory.

6. The Admin Command Center (adminpage)

The Admin page is split into two power-zones:

Zone A: The 4-Type Notification Panel The Admin monitors and approves four specific types of requests:

1. **New Account Creation:** Verifying and activating new faculty.
2. **Paper Deletions:** Approving the removal of research from the repository.
3. **Department Change:** Confirming a faculty member's move to a new department.
4. **Position Change:** Validating promotions or title changes.

Zone B: Analytical & Global Oversight

- **All Profile Details:** A master table with a "Search & Toggle" feature to manage every faculty member's status (Active/Inactive).
- **Analytical Tables:** Two specialized tables used for institutional reporting—tracking which departments are hitting their research targets and identifying the growth trends of the college.